Conditioning of random cyclic-shift Krylov compressions

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Independently reviewed resolution. Authorship is recorded at the submitter's request. The supplied AI-generated proof and computational support are disclosed in the submission record. The exact canonical target is resolved. The dated independent agent review records the subsequent checks and supersedes original draft remarks about pending review. This is not external human peer review or formal certification.

1. Claim and scope

This manuscript proposes an affirmative resolution of repository problem **IE-10**, with exactly its complex-sphere starting-vector model.[1] The original workshop question specifically mentions the cyclic shift; a counterexample for general nonnormal inputs does not address this case.[2]

Theorem 1 (cyclic shift). Let $n \geq 3$, let $2 \leq k < n$, and let C_n be the cyclic shift on $\mathbb{C}^n$. Draw b uniformly from the complex unit sphere. Let Q have orthonormal columns spanning

$$\operatorname{span}\{b, C_n b, \dots, C_n^{k-1} b\}, \qquad H_k = Q^* C_n Q.$$

For a diagonalizable matrix define

$$\kappa_V(H) = \inf_{H = VDV^{-1},\ D \text{ diagonal}} \|V\|_2 \|V^{-1}\|_2,$$

and set this quantity to infinity otherwise. Then

$$\mathbb{E}\, \kappa_V(H_k) \leq 17 n^2 k. \tag{1}$$

Consequently, for every $0 < \eta < 1$,

$$\Pr\left\{ \kappa_V(H_k) \leq \frac{17}{\eta} n^2 k \right\} \geq 1 - \eta. \tag{2}$$

In particular, $C = 1700$ **and** $c = 3$ satisfy the repository's uniform 0.99 target.

[1] Source 1, the displayed probability target and starting-vector model.
[2] Source 2, Section 3.3, Problem 3.5 and its 20 August 2026 update.

Two probability arguments are given. Section 5 already proves the repository's existence claim, with the weaker constants $C = 2{,}000{,}000$ and $c = 5$, using discriminant avoidance and Cauchy's estimate. Sections 6–7 improve this to (1)–(2) by controlling the total variation of a rank-one root locus. Thus the affirmative answer does not rely solely on the sharper root-locus estimate.

No assertion is made for real-sphere starts, arbitrary nonnormal input matrices, or finite-precision construction of an eigenbasis.

2. Exact weighted-polynomial model

We work first with arbitrary distinct points $\zeta_1, \dots, \zeta_n$ on the unit circle. Write

$$d = \min_{i \neq j} |\zeta_i - \zeta_j| > 0,$$

and let $w_1, \dots, w_n$ be independent exponential random variables of mean one. On the sample space $\mathbb{C}^{\{\zeta_1,\dots,\zeta_n\}}$, use

$$\langle f, g \rangle_w = \sum_{j=1}^{n} w_j \overline{f(\zeta_j)} g(\zeta_j).$$

Let $\mathcal{P}_{k-1}$ be the evaluated polynomials of degree less than k, let P_w be its orthogonal projection, and set

$$H_w = P_w M_z\big|_{\mathcal{P}_{k-1}}, \qquad (M_z f)(\zeta_j) = \zeta_j f(\zeta_j).$$

Positive weights and distinct nodes make the evaluation map injective on this polynomial space. Since M_z is unitary, H_w is a contraction.

The general estimate proved below is

$$\mathbb{E}\, \kappa_V(H_w) \le 2k + \frac{64nk}{d}. \tag{3}$$

Here and below an operator condition number is measured in its stated Hilbert-space norm, equivalently in any orthonormal coordinates.

To obtain the cyclic model, diagonalize C_n by a unitary Fourier matrix. A uniformly distributed complex unit vector can be represented as $g/\|g\|_2$, where the coordinates of g are independent standard complex Gaussians. This follows directly from the unitary invariance of the density proportional to $\exp(-\|g\|_2^2)$. In polar coordinates, $|g_j|^2$ are independent mean-one exponentials. The Krylov vectors in spectral coordinates are the evaluations of $1, z, \dots, z^{k-1}$ multiplied coordinatewise by $g_j/\|g\|_2$. Their phases are removed by a diagonal unitary commuting with the spectral matrix. The common normalization of the weights does not change an orthogonal compression. Thus H_k is unitarily equivalent to H_w with the nth roots of unity as nodes. This also proves the almost-sure dimension assertion.

For coefficient calculations put

$$a_j = (1, \zeta_j, \dots, \zeta_j^{k-1}), \qquad G = \sum_j w_j a_j^* a_j, \qquad F = \sum_j w_j \zeta_j a_j^* a_j.$$

The coefficient representation of H_w is $G^{-1}F$, with coefficient inner product $x^* G y$.

3. Conditioning is controlled by weight sensitivity

3.1 A conjugation that matches left and right coordinate magnitudes

Define an antilinear map on the full sample space by

$$(Jf)(\zeta_j) = \zeta_j^{k-1}\overline{f(\zeta_j)}.$$

It is an antiunitary involution. For a polynomial of degree less than k, it reverses and conjugates the coefficients, so it preserves $\mathcal{P}_{k-1}$. Hence J commutes with P_w. Directly,

$$JM_zJ = M_z^*, \qquad JH_wJ = H_w^*. \tag{4}$$

Suppose λ_i is simple and r_i is a unit right eigenvector of H_w. Then $\ell_i = Jr_i$ is a unit left eigenvector, meaning $H_w^*\ell_i = \overline{\lambda_i}\ell_i$. Put

$$\alpha_i = \langle \ell_i, r_i\rangle_w, \qquad \kappa_i = \frac{1}{|\alpha_i|}, \qquad p_{ij} = w_j|r_i(\zeta_j)|^2.$$

Simplicity gives $\alpha_i \neq 0$. The number κ_i is the norm of the rank-one spectral projector. Also

$$\sum_j p_{ij} = 1, \qquad w_j|\ell_i(\zeta_j)|^2 = p_{ij}. \tag{5}$$

3.2 An exact derivative identity

Differentiate the generalized eigenvalue equation $Fr_i = \lambda_i G r_i$. Multiplication by the generalized left eigenvector cancels the derivative of r_i, giving

$$\frac{\partial\lambda_i}{\partial w_j} = \frac{(\zeta_j - \lambda_i)\overline{\ell_i(\zeta_j)}r_i(\zeta_j)}{\alpha_i}.$$

Therefore

$$\sum_j w_j\left|\frac{\partial\lambda_i}{\partial w_j}\right| = \kappa_i\sum_j p_{ij}|\zeta_j - \lambda_i|. \tag{6}$$

This identity is taken at simple-spectrum weight vectors; those vectors have full measure by Section 4.

Lemma 2 (deterministic sensitivity bound). At every positive simple-spectrum weight vector,

$$\kappa_i \le \max\left\{2, \frac{8}{d}\sum_j w_j\left|\frac{\partial\lambda_i}{\partial w_j}\right|\right\} \le 2 + \frac{8}{d}\sum_j w_j\left|\frac{\partial\lambda_i}{\partial w_j}\right|. \tag{7}$$

Proof. Each summand in $\alpha_i = \sum_j w_j\overline{\ell_i(\zeta_j)}r_i(\zeta_j)$ has modulus p_{ij}. The reverse triangle inequality implies

$$|\alpha_i| \ge 2\max_j p_{ij} - 1.$$

If $\kappa_i > 2$, then $\max_j p_{ij} < 3/4$. At most one node can lie at distance strictly less than $d/2$ from λ_i. Consequently at least one quarter of the probability mass in (5) is at distance at least $d/2$, and

$$\sum_j p_{ij}|\zeta_j - \lambda_i| \geq d/8.$$

Now use (6). The case $\kappa_i \leq 2$ is immediate. □

3.3 From individual projectors to an eigenbasis

Lemma 3 (balanced column scaling). For any matrix with simple spectrum,

$$\kappa_V(H) \leq \sum_{i=1}^{k} \kappa_i, \tag{8}$$

where the κ_i are the norms of its rank-one spectral projectors.

Proof. Use orthonormal ambient coordinates. Let R have unit right eigenvectors as columns. The ith row of R^{-1} has norm κ_i: it is the dual left eigenvector with pairing one. Define $D = \operatorname{diag}(\sqrt{\kappa_1}, \dots, \sqrt{\kappa_k})$ and $V = RD$. Then

$$\|V\|_F^2 = \sum_i \kappa_i, \qquad \|V^{-1}\|_F^2 = \sum_i \frac{\kappa_i^2}{\kappa_i} = \sum_i \kappa_i.$$

The operator norms are bounded by the Frobenius norms, and V is still an eigenbasis. □

4. Simplicity and rank-one conditional pencils

The discriminant in z of

$$p(z, w) = \det(zG(w) - F(w))$$

is a polynomial in the weights and is not identically zero. To see this, allow boundary weights supported positively on exactly k distinct nodes. The resulting G remains positive definite. The square evaluation matrix on those nodes is invertible, and $G^{-1}F$ is similar to the diagonal matrix of those k distinct nodes. Its discriminant is nonzero. Thus the discriminant polynomial is nonzero, and its zero set in real weight space has measure zero.

Fix an index j and condition on the other weights. For almost every such conditioning the discriminant, considered as a polynomial in $t = w_j$, is not identically zero: at least one of its coefficient polynomials in the remaining weights is nonzero.

Write $a = a_j$ and $\zeta = \zeta_j$. The conditional pencil is

$$G(t) = G_0 + ta^*a, \qquad F(t) = F_0 + t\zeta a^*a.$$

Because $k < n$, the other $n-1$ positive-weight nodes span the polynomial space, so $G_0 \succ 0$. By the rank-one determinant identity, or multilinearity of the determinant,

$$p(z, t) = \det(zG(t) - F(t)) = p_0(z) + tp_1(z), \qquad \deg p_0, \deg p_1 \leq k. \tag{9}$$

Its leading coefficient in z is $\det G(t) > 0$ for real $t > 0$. Its discriminant $\Delta(t)$ has degree at most $2k-2$, since the discriminant is homogeneous of that degree in the coefficients of a degree-k polynomial and each coefficient in (9) is affine in t.

For positive real t, every root lies in the closed unit disk. Except at the finitely many positive zeros of Δ, the roots admit locally analytic labels. All sums over the roots used below are independent of the chosen labeling.

5. A direct probability proof using discriminant avoidance

This section independently proves a polynomial bound sufficient for IE-10, without the length estimate of Section 6.

Proposition 4 (weaker uniform tail bound). In the cyclic model, for $0 < \eta < 1$,

$$\Pr\left\{\kappa_V(H_k) \le \frac{192}{\eta^2} n^3 k^2\right\} \ge 1 - \eta. \tag{10}$$

In particular, $C = 2{,}000{,}000$, $c = 5$ work at success probability 0.99.

Proof. Put $\varepsilon = \eta/(16nk)$. For one conditional pencil, call a positive real t_0 bad if $t_0 \le 2\varepsilon$ or if it is within distance 2ε of a complex zero of Δ. There are at most $2k-2$ such zeros, counted with multiplicity. A disk of radius 2ε cuts the real line in an interval of length at most 4ε. Since the exponential density is bounded by one, the conditional bad probability is at most

$$2\varepsilon + 4(2k-2)\varepsilon \le 8k\varepsilon.$$

The exceptional conditionings from Section 4 have probability zero. Union over the n coordinates therefore costs at most $8nk\varepsilon = \eta/2$.

At a good t_0, every root is holomorphic on $|t - t_0| < \varepsilon$. Indeed, the discriminant does not vanish there, and $G(t)$ is invertible in $\Re t > 0$. For an eigenpair of the complex conditional pencil, set

$$g = x^* G_0 x > 0, \quad m = x^* F_0 x, \quad s = |ax|^2 \ge 0.$$

The positive-weight representation of F_0 gives $|m| \le g$. Multiplying the generalized eigenvalue equation by x^* gives

$$z = \frac{m + t\zeta s}{g + ts}.$$

Throughout the disk, $\Re t > t_0/2$ and $|t| < 3t_0/2$, so

$$|z| \le \frac{g + (3t_0/2)s}{g + (t_0/2)s} \le 3.$$

Cauchy's estimate yields, for every eigenvalue and every good coordinate,

$$\left|\frac{\partial \lambda_i}{\partial w_j}\right| \le 3/\varepsilon. \tag{11}$$

Let $W = \sum_j w_j$. Since $\mathbb{E}W = n$, Markov's inequality gives

$$\Pr\{W > 2n/\eta\} \leq \eta/2.$$

On the joint good event, of probability at least $1 - \eta$, equations (7), (8), and (11) give

$$\kappa_V(H_k) \leq k \max\left\{2, \frac{24W}{d\varepsilon}\right\} \leq \frac{768n^2k^2}{d\eta^2}.$$

For cyclic nodes, $d \geq 4/n$; the bound dominates $2k$ and proves (10). $\square$

6. Total variation of the conditional root locus

Lemma 5 (bounded root-locus length). For almost every conditioning in Section 4,

$$\sum_{i=1}^{k} \int_0^\infty |\lambda_i'(t)|\, dt \leq 8k, \tag{12}$$

where the integrals are taken over the simple-spectrum intervals. Values at the finite collision set are irrelevant.

Proof. If p_1 vanishes identically, all roots are constant. Otherwise define the real polynomial

$$q(x, y) = \Im\left(p_0(x+iy)\overline{p_1(x+iy)}\right), \qquad \deg q \leq 2k.$$

A moving root, away from a common root of p_0 and p_1, satisfies

$$t = -\frac{p_0(z)}{p_1(z)} \in (0, \infty), \qquad q(\Re z, \Im z) = 0. \tag{13}$$

Common roots give constant branches. A moving branch can meet one only at an exceptional collision parameter, so these points do not affect the integrals.

If q vanishes identically, the holomorphic rational function p_0/p_1 is real valued on its domain and is therefore constant. Equation (9), together with the nonvanishing leading coefficient for positive t, again implies that all roots are constant. We may therefore assume $q \not\equiv 0$.

For all but finitely many real u, the polynomial $q(u, y)$ is nonzero and has at most $2k$ real roots. Each moving-root point $u+iy$ with $p_1(u+iy) \neq 0$ determines at most one parameter t through (13), and at a simple-spectrum parameter it belongs to only one root. Consequently, for almost every u, the number of moving-root crossings of the vertical line $\Re z = u$, counted with their parameter preimages, is at most $2k$. The same statement holds for horizontal lines.

For clarity, this crossing count bounds variation, not merely the set-theoretic image of a curve. On each simple-spectrum interval a root is analytic. Split each real or imaginary coordinate into its monotonicity intervals, ignoring constant coordinates, and apply one-dimensional change of variables on those intervals. Countably many intervals suffice; nonnegative integrands allow summation by Tonelli's theorem. Since all roots are in the closed unit disk, their real and imaginary coordinates are in $[-1,1]$. Thus

$$\sum_i \int_0^\infty |\Re\lambda_i'(t)|\,dt \le \int_{-1}^1 2k\,du = 4k,$$

and likewise

$$\sum_i \int_0^\infty |\Im\lambda_i'(t)|\,dt \le 4k.$$

The exceptional coordinate values corresponding to collisions, constant branches, or lines contained in $q=0$ form a null set. Finally, $|z'| \le |\Re z'| + |\Im z'|$ gives (12). $\square$

7. Proof of the expectation bound

Condition on all weights except $w_j = t$. Its remaining density is e^{-t} on $(0,\infty)$. Lemma 5 gives

$$\begin{aligned}\mathbb{E}\left[\sum_i w_j \left|\frac{\partial\lambda_i}{\partial w_j}\right| \,\middle|\, (w_\ell)_{\ell\neq j}\right] &= \int_0^\infty te^{-t}\sum_i |\lambda_i'(t)|\,dt \\ &\le 8k,\end{aligned} \tag{14}$$

where we used only $te^{-t} \le 1$. Exceptional conditionings have probability zero. Sum (14) over j, and then use (7)–(8) and Tonelli's theorem:

$$\mathbb{E}\,\kappa_V(H_w) \le 2k + \frac{8}{d}\mathbb{E}\sum_{i,j} w_j \left|\frac{\partial\lambda_i}{\partial w_j}\right| \le 2k + \frac{64nk}{d}.$$

This proves (3). For the cyclic shift,

$$d = 2\sin(\pi/n) \ge \frac{4}{n},$$

using concavity of sine on $[0,\pi/2]$. Therefore

$$\mathbb{E}\,\kappa_V(H_k) \le 2k + 16n^2k \le 17n^2k.$$

Markov's inequality proves Theorem 1.

Corollary 6 (one starting vector, all dimensions). With the same random b used for every k, no independence between the compressions is needed to conclude

$$\Pr\left\{\kappa_V(H_k) \le \frac{17}{2\eta}n^4 \text{ for every } 2 \le k < n\right\} \ge 1-\eta.$$

Indeed, the expectation of the sum of their condition numbers is at most $17n^2\sum_{k=2}^{n-1} k \le (17/2)n^4$.

8. Checks, limitations, and provenance

The accompanying code checks the weighted model against an orthonormal Krylov construction, the matched left/right magnitudes, the eigenvalue derivative identity, balanced column scaling, finite-difference derivatives, and sampled conditional root paths. A separate 140-decimal-digit diagnostic includes weight ratios of 10^{24}, closely clustered nodes, and cyclic weights within 10^{-35} of the defective uniform example. These stress tests are numerical, not interval certificates. An exact $n = 3$, $k = 2$ example with weights $(t, 1, 1)$ gives

$$G = \begin{pmatrix} t+2 & t-1 \\ t-1 & t+2 \end{pmatrix}, \qquad F = \begin{pmatrix} t-1 & t-1 \\ t+2 & t-1 \end{pmatrix},$$

$$\det(zG - F) = 3\big((2t+1)z^2 + (1-t)z + (1-t)\big),$$

with discriminant $81(t-1)(t+1/3)$. This illustrates why an almost-sure argument must allow exceptional defective weights; the point $t = 1$ is not silently excluded by an unproved uniform deterministic bound.

This example also permits an exact geometric check of Lemma 5. For $0 < t < 1$, the roots lie on the circle $|z+1| = 1$ and each moves monotonically from angle $\pm\pi/3$ to angle zero about its center. For $t > 1$, one root moves from zero to 1 and the other from zero to $-1/2$. Indeed,

$$\frac{dt}{dz} = \frac{3z(z+2)}{(2z^2 - z - 1)^2},$$

which has the required sign on each real interval. The total root-path length is therefore $2\pi/3 + 3/2 < 16 = 8k$. This is a check on one explicit pencil, not a substitute for the general crossing-count proof.

The tests are diagnostics and exact algebraic checks, not an independent proof of the probability theorem. The proof uses randomness only in the starting vector through independent exponential spectral weights. It never substitutes an independent Haar subspace for the Krylov subspace. Both proofs retain the restrictions $k < n$ and complex-sphere randomness.

This is an AI-generated proposed argument, prepared for mathematical review on 11 September 2026. It has not been independently refereed or formally verified. No assertion of priority is made. The proposed repository status is **Solution claimed**, not independently established **Solved**.

Sources

1. OpenProblemsInNLA. IE-10: Conditioning of a random Krylov compression of a cyclic shift. Problem statement and audit update of 10 September 2026. Accessed 11 September 2026.
2. Noah Amsel et al. *Linear Systems and Eigenvalue Problems: Open Questions from a Simons Workshop.* arXiv:2602.05394v3, 21 August 2026. Section 3.3, Problem 3.5 and its update. This source identifies the question and scope; it is not the source of the proposed proof.